Introduction to Active Directory and Account Management

Chapter 4

Objectives

- Understand Active Directory (AD) basic concepts
- Install and configure Active Directory
- Implement Active Directory containers
- Create and manage user accounts
- Configure and use security groups
- Understand user profiles
- Describe important additional features in Active Directory

Active Directory Basics

- Active Directory (AD Database)
  - Directory service that houses information about all network resources such as servers, printers, user accounts, groups of user accounts, security policies, and other information
- Directory service
  - Responsible for providing a central listing of resources and ways to quickly find and access specific resources and for providing a way to manage network resources
- Also referred to as Active Directory Domain Services (ADDS)
  - This is the name of the role you will install

Active Directory Basics

- Domain controllers (DCs)
  - Servers that have the AD DS server role installed
  - Contain writable copies of information in Active Directory
  - Recommended a minimum of two DCs in any organization
- Member servers
  - Servers on a network managed by Active Directory that do not have Active Directory installed
- Domain
  - Container that holds information about all network resources that are grouped within it
  - Every resource is called an object

Active Directory Basics

- Active Directory Basics
  - **Active Directory Objects**
    - User Accounts, Groups, Computers, Printers, Shares, Organization Units, and many more.
    - If an object contains nothing within it, it is referred to as Leaf Objects or a Terminal Object (they are at the end of a branching hierarchy).
    - If an Object does contain other objects, it is called a container object.
    - All objects and their properties are defined in the Active Directory Schema.
  - **Querying objects in the Directory**
    - Lightweight directory Access Protocol (LDAP) is used to query (search/find) objects.
    - LDAP uses a naming syntax called Distinguished Names (DN)

Do this In-Class Activity

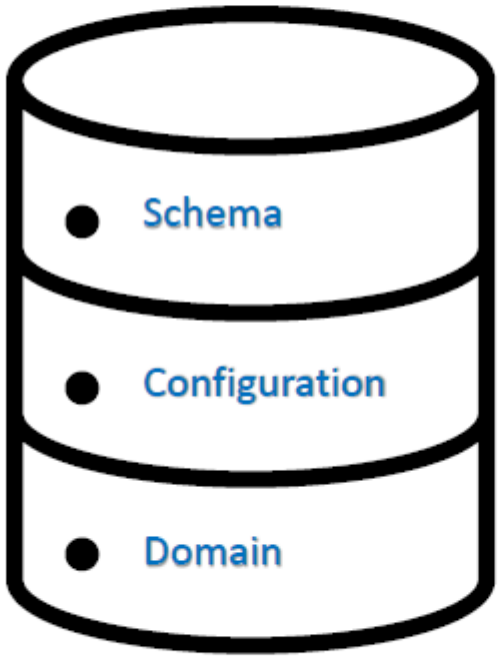
- Install Active Directory Domain Services Role
  - Follow along…

Active Directory Basics

- Multimaster replication
  - Each DC is equal to every other DC in that it contains the full range of information that composes Active Directory
  - If information on one DC changes it is replicated to all other DCs
- Active Directory is built to make replication efficient
- Active Directory in Windows Server 2022 can:
  - Replicate individual properties instead of entire accounts
  - Replicate Active Directory based on the speed of the network link
- Three general concepts important for understanding Active Directory:
  - Schema
  - Global catalog
  - Namespace

Active Directory Basics

Active Directory Databased Partitions



I describe all objects and their properties.

I determine the structure and layout of the domain.

I contain all the objects that are described above.

Global Catalog

I am part of the Configuration Partition. I am an index of the objects in each domain.

Schema

- Active Directory schema
  - Defines *the objects* that can exist in Active Directory
  - Defines *the attributes of objects* that can exist in Active Directory
  - Like a small database of information associated with that object
  - The is ONE schema per *forest*
- Schema information for objects in a domain is replicated on every DC
- User account
  - One class of object in Active Directory that is defined through schema elements unique to that class
  - Example: Which ‘tabs’ exist, and what is in the ‘tabs’ of a user account, is governed by the schema.

Schema

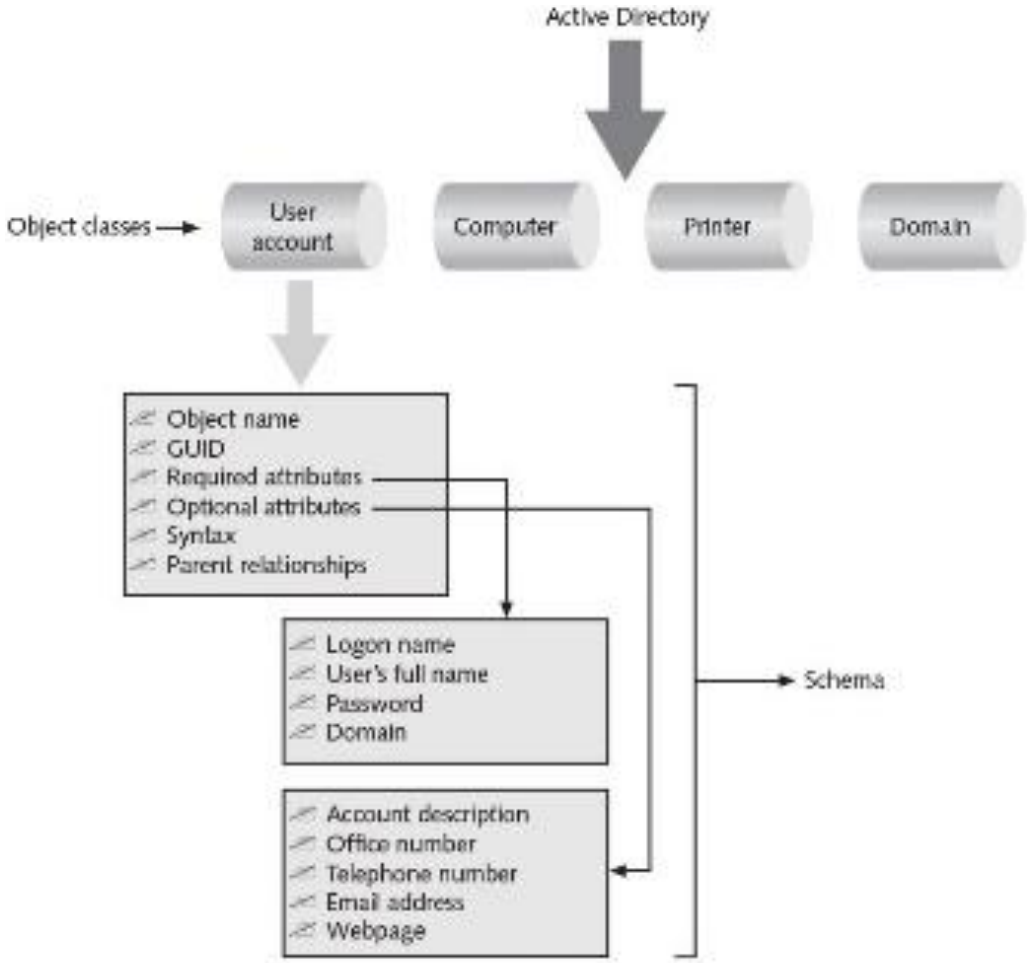


Figure 4-8 Sample schema information for user accounts

Global Catalog

- Global catalog
  - Stores information about every object within a forest
  - The global catalog (GC) allows users and applications to find objects in an Active Directory domain tree, given one or more attributes of the target object.
- Global catalog server
  - Store a *full* replica of every object within its own domain and a *partial* replica of each object within every domain in the forest, but only a small number of attributes for each object (most frequently used).
- The GC server enables forest-wide searches of data
- The GC allows users to quickly find objects in the directory.

Global Catalog (GC)

- The global catalog serves the following purposes:
  - Serving as the central storehouse of **key object** information in a forest that has multiple domains
  - Providing lookup and access to all resources in all domains
  - Providing replication of key Active Directory elements
  - Keeps a copy of the most used attributes for each object for quick access
- By default, the first DC in the forest is automatically designated as the global catalog server
  - You have the option of configuring another DC to be a global catalog server as well as designating multiple DCs as global catalog servers
  - While the GC is built automatically, the attributes stored in the GC can be customized.

Show Servers Holding FSMO Roles (Powershell)

SHOW GC SERVERS

```
# Import the module if not already loaded (usually auto-loads)
Import-Module ActiveDirectory

# Show all Global Catalog servers
Get-ADForest | Select-Object -ExpandProperty GlobalCatalogs | Sort-Object |
Format-Table -AutoSize
```

SHOW FSMO ROLES/ROLE HOLDERS

```
# Forest-wide roles (Schema Master & Domain Naming Master)
Get-ADForest | Select-Object SchemaMaster, DomainNamingMaster | Format-List

# Domain-wide roles (PDC Emulator, RID Master, Infrastructure Master)
Get-ADDomain | Select-Object PDCEmulator, RIDMaster, InfrastructureMaster |
Format-List
```

Namespace

- Active Directory uses Domain Name System (DNS)
  - There must be at least one DNS server on the network that Active Directory can access
- DNS
  - A TCP/IP-based name service that converts computer and domain host names to IP addresses and vice versa, through a process called name resolution
- Namespace
  - A logical area on a network that contains directory services and named objects
  - Has the ability to perform name resolution
- Active Directory employs two kinds of namespaces:
  - Contiguous – one in which every child object contains the name of the parent object
  - Disjointed – when the child's name does not resemble the name of its parent object

Containers in Active Directory

- Active Directory has a treelike structure
- The hierarchical elements, or containers, of Active Directory include forests, trees, domains, organizational units (OUs), and sites

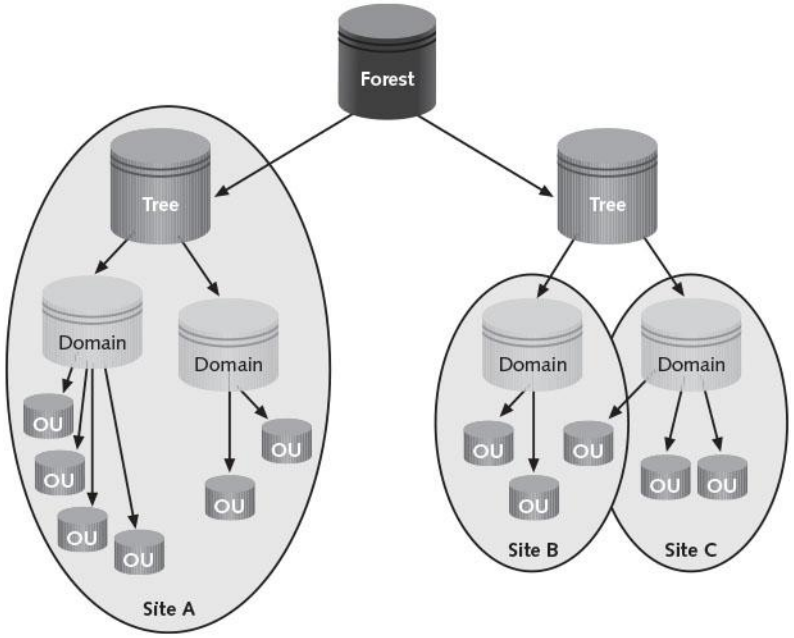


Figure 4-9 Active Directory hierarchical containers

Forest

- Forest
  - Consists of one or more Active Directory trees that are in a common relationship
  - The forest root domain can also be part of a tree in the forest.
- Forests have the following characteristics:
  - A set of one or more domain trees that do not form a contiguous namespace.
  - Unlike trees, a forest does not require a distinct namespace.
  - All trees use the same schema
  - All trees use the same global catalog
  - Domains enable administration of commonly associated objects, such as accounts and other resources, within a forest
  - Two-way transitive trusts are automatically configured between domains within a single forest

Forest

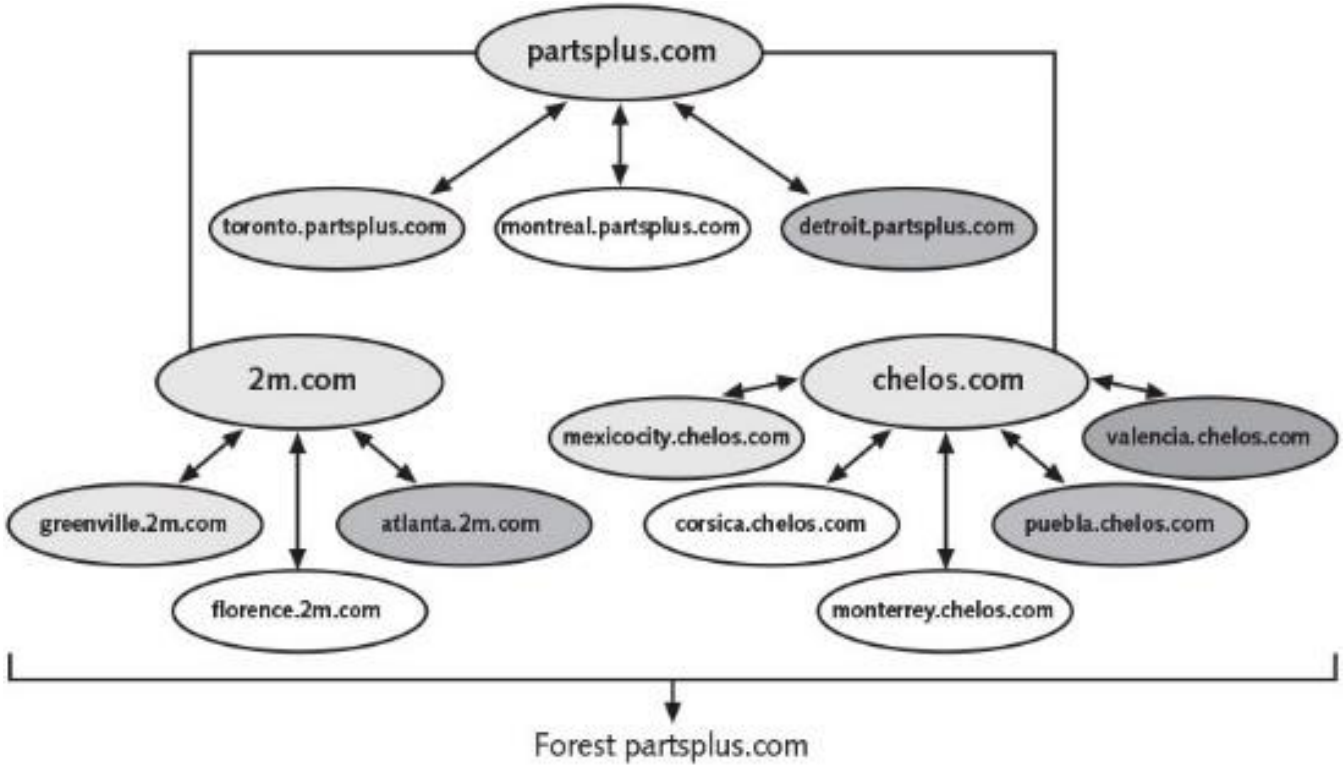


Figure 4-10 Forest partsplus.com

Forest

- A forest provides a means to relate trees that use a contiguous namespace in domains within each tree
  - But that have disjointed namespaces in relationship to each other
- The advantage of joining trees into a forest is that all domains share the same schema and global catalog
- Forest functional level
  - Refers to the Active Directory functions supported forest-wide

Forest

- Windows Server 2022 Active Directory recognizes several types of forest functional levels:
  - Windows 2000 Native Forest Functional Level
  - Windows Server 2003 Forest Functional Level
  - Windows Server 2008 Forest Functional Level
  - Windows Server 2008 R2 Forest Functional Level
  - Windows Server 2012 Forest Functional Level
  - Windows Server 2012 R2 Forest Functional Level
  - Windows Server 2016 Forest Function Level
- When servers are upgraded, it might make sense to raise the forest functional level to match the server OSs in use.
- You cannot (easily) down grade forest functional level.

Forest Functional Levels

Table 4-3 Windows Server 2019 forest functional levels

Functional level	New Active Directory features provided
Windows Server 2008	No additional features beyond those within the Windows Server 2008 domain functional level
Windows Server 2008 R2	The ability to create and use the **Active Directory Recycle Bin** to recover deleted objects
Windows Server 2012	No additional features beyond those within the Windows Server 2012 domain functional level
Windows Server 2012 R2	No additional features beyond those within the Windows Server 2012 R2 domain functional level
Windows Server 2016	The ability to use the **Microsoft Identity Manager (MIM)** to restrict malicious access to Active Directory using **Privilege Access Management (PAM)**

Tree

- Tree
  - Contains one or more domains that are in a common relationship
- Tree has the following characteristics:
  - Domains are represented in a contiguous namespace and can be in a hierarchy
  - Two-way trust relationships exist between parent domains and child domains
  - All domains in a single tree use the same schema for all types of common objects
  - All domains use the same global catalog
  - Child domains share a portion of their namespace with the parent.

Tree

- The domains in a tree typically have a hierarchical structure
  - Such as a root domain at the top and other (child) domains under the root
- The domains within a tree are in what is called a Kerberos transitive trust relationship
  - Which consists of two-way trusts between parent domains and child domains
- Because of the trust relationship between parent and child domains, any one domain can have access to the resources of all others

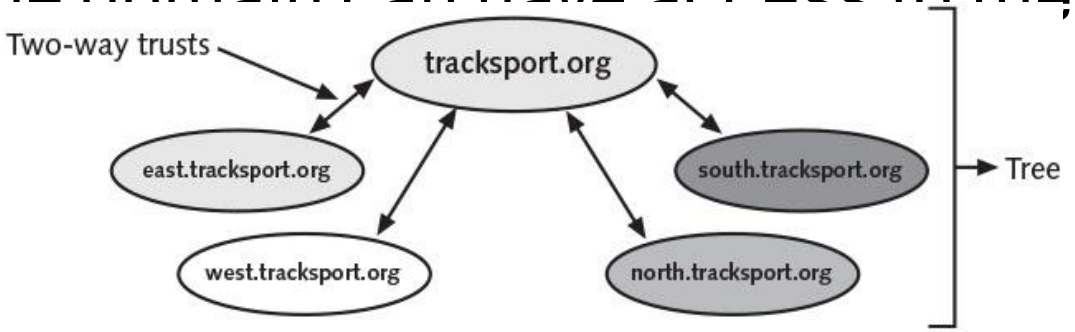


Figure 4-12 Tree with hierarchical domains

Domain

- Microsoft views a domain as a logical partition within an Active Directory forest
  - A domain is a grouping of objects that typically exists as a primary container within Active Directory
- The basic functions of a domain are as follows:
  - To provide an Active Directory “partition’’ in which to house objects that have a common relationship, particularly in terms of management and security
  - To establish a set of information to be replicated from one DC to another
  - To expedite management of a set of objects
- As a best practice, domains (and forests) should be as ‘flat’ and simple as is feasible. The ideal scenario is one domain|one forest.

Domain

- Domain functional levels
  - Refers to the Windows Server operating systems on domain controllers and the domain-specific functions they support
- The domain functional levels are as follows:
  - Windows 2000 Domain Functional Level
  - Windows Server 2003 Domain Functional Level
  - Windows Server 2008 Domain Functional Level
  - Windows Server 2008 R2 Domain Functional Level
  - Windows Server 2012 Domain Functional Level
  - Windows Server 2012 R2 Domain Functional Level
  - Windows Server 2016 Domain Functional Level
- You cannot (easily) down grade Domain functional level.

Domain Functional Levels

Table 4-2 Windows Server 2019 domain functional levels

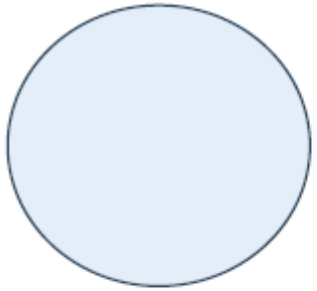
Functional level	New Active Directory features provided
Windows Server 2008	**Distributed File System (DFS)** replication between domain controllers, **Advanced Encryption Standard (AES)** security for Kerberos authentication, and enhanced user account password policies
Windows Server 2008 R2	**Service Principle Name (SPN)** identification for network services
Windows Server 2012	**Compound authentication**, which creates Kerberos tickets with additional information used by other services, and **Kerberos armoring**, which creates a secure channel for authentication that is protected against network attacks
Windows Server 2012 R2	Additional encryption technologies for Kerberos authentication, as well as the ability to create authentication policies
Windows Server 2016	Additional Kerberos authentication features

Organizational Unit

- Organizational unit (OU)
  - a container within a Microsoft Active Directory domain which can hold users, groups and computers.
  - Smallest unit of organization in an AD Forest.
  - Offers a way to achieve more flexibility in managing the resources associated with a business unit, department, or division (or any other form of organization) than is possible through domain administration alone
- An OU is a grouping of related objects within a domain
  - OUs allow the grouping of objects so that they can be administered using the same group policies
- OUs can be nested within OUs

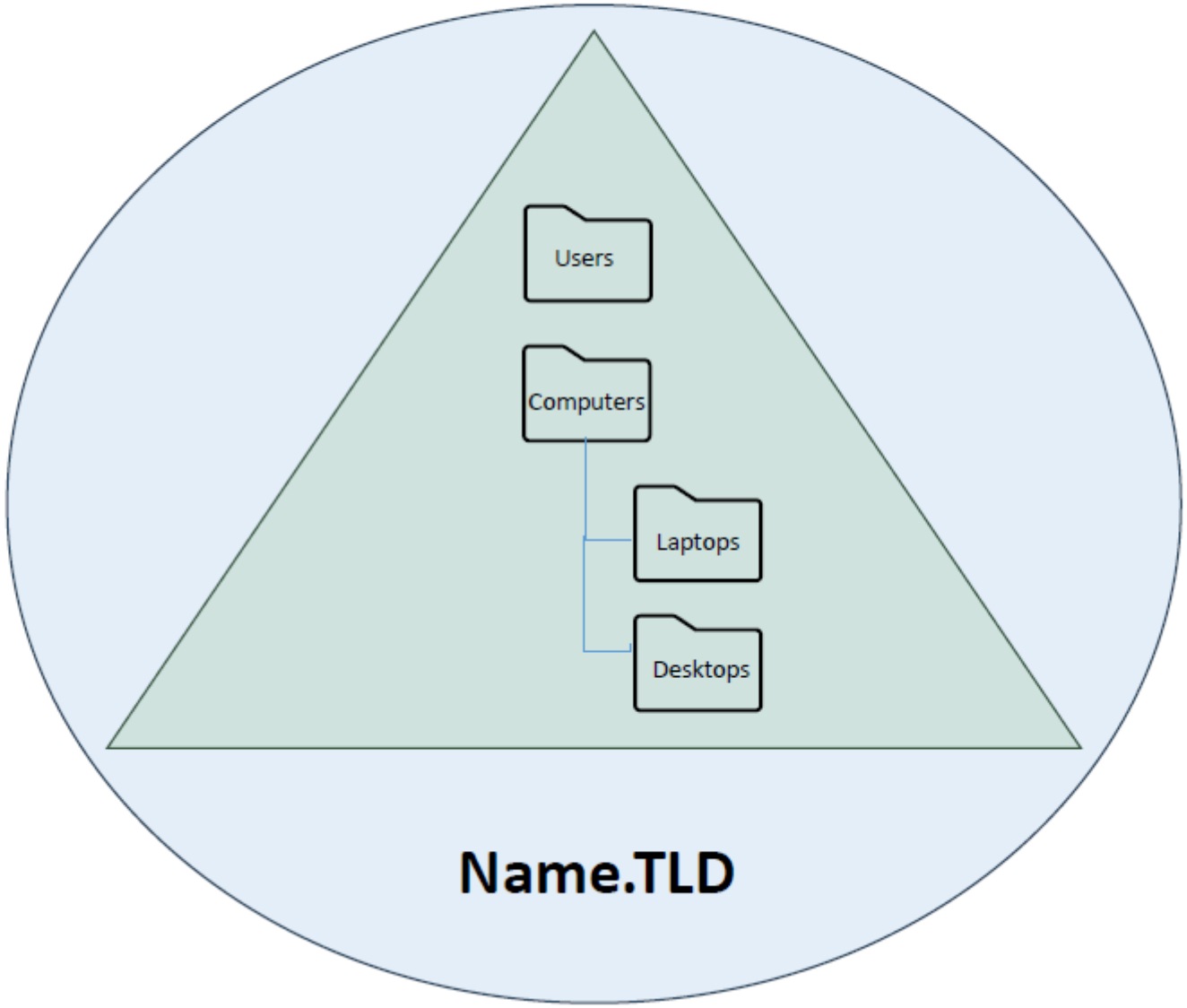
Active Directory Basics

Domain or
Child/Domain



Forest

Organizational Unit



Organizational Unit

- When you plan to create OUs, keep three concerns in mind:
  - Microsoft recommends that you limit OUs to 10 levels or fewer
  - Active Directory works more efficiently when OUs are set up horizontally instead of vertically
  - The creation of OUs involves more processing resources because each request through an OU requires CPU time

Organizational Unit

- Vertical OU Structure (deep/nested hierarchy)
  - This creates many levels of nested OUs, often mirroring the company's strict organizational chart (e.g., Continent → Country → Region → City → Department → Team → Sub-team).
  - Example:

```
Domain
└── Europe
    └── Germany
        └── Berlin
            └── Sales
                └── Team A
                    └── Users
```

Organizational Unit

- Horizontal OU Structure (shallow/flatter hierarchy)
  - This keeps nesting minimal (usually 2–4 levels max) and groups objects more broadly by administrative need, policy requirements, or function rather than strict org-chart depth.
  - Example:

```
Domain
├── Users
│   ├── All-Users (broad policies)
│   └── Sales-Users (specific sales policies)
├── Computers
│   ├── Workstations
│   └── Servers
└── Locations (if needed for site-specific policies)
    └── Berlin-Computers
```

Site

- Site
  - Sites are physical groupings of IP subnets
  - used to replicate information among domain controllers (DC) efficiently.
  - A TCP/IP-based concept (container) within Active Directory that is linked to IP subnets
- A site has the following functions:
  - Reflects one or more interconnected subnets
  - Reflects the physical aspect of the network
  - Is used for DC replication
  - Is used to enable a client to access the DC that is physically closest
  - Is composed of only two types of objects, servers and configuration objects
- Sites are based on connectivity and replication functions
- A site may be single or multi-domain

Sites

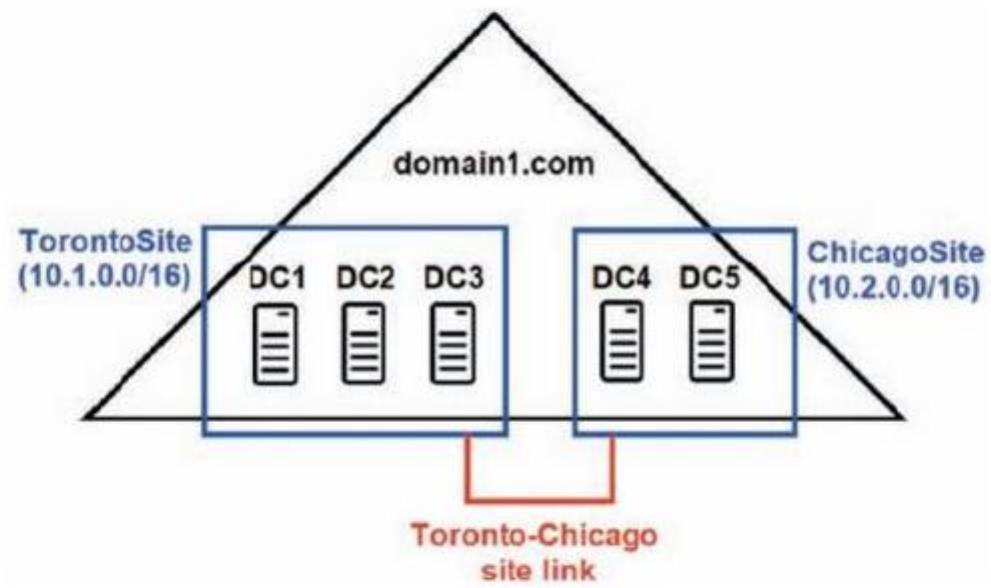


Figure 4-10 A domain with five domain controllers within two sites

Active Directory Guidelines

- Above all, keep Active Directory as simple as possible
  - Plan its structure before you implement it
- Implement the least number of domains possible
  - With one domain being the ideal and building from there
- Implement only one domain on most small networks
- Use OUs to reflect the organization's structure
- Create only the number of OUs that are absolutely necessary
- Do not build an Active Directory with more than 10 levels of OUs

Active Directory Guidelines

- Use domains as *partitions* in forests to demarcate commonly associated accounts and resources governed by group and security policies
- Implement multiple trees and forests only as necessary
- Use sites in situations where there are multiple IP subnets and multiple geographic locations (to improve logon and DC replication performance)

ICA: Do this…

- Promote server to a Domain Controller
  - Follow along…

Azure Active Directory

- Microsoft Azure Active Directory
  - An Active Directory service that an organization can use for online cloud applications such as Office 365
  - Provides user authorization and identity management for organizations that subscribe to Microsoft cloud services
- Cloud computing
  - Involves providing a host of scalable web-based applications and services over the Internet or a private network that are used by clients through web browsers and downloadable apps

Azure AD: what does it do?

- Stores users and groups and can be used for cloud-based authentication.
- All MS cloud solutions (office 365, Azure, etc) use Azure AD.
- Manages application access and integrates with other third-party cloud applications (Salesforce, Slack, etc).

Azure AD: What it *doesn't* do.

- Cannot join member servers
- Can join Windows 10 clients, but a bit differently.
- No group Policy,
- No LDAP, Kerberos or NTLM.
- Flat directory structure (no OUs, no forests)

- Azure AD is not *replacement* for AD.

AD + Azure AD

- Regular Active Directory is used to manage on-prem infrastructure
- It can be paired with Azure AD to provide access to cloud Applications and provide web-based logon services (Microsoft Entra Connect)

Organizational Units

- The building blocks of Active Directory
- Thorough planning of the OU structure eases managing users and computers and applying group policies
- Easily rearranged to reflect changing organizational needs
- OU's may contain leaf objects or other OUs

Organizational Units

- OU's primary purpose is:
  - To serve as a point to apply group policy objects to the users and computers within them
  - To server as a point to delegate control that allows one user to exercise administrative control over the users within the OU

Naming Conventions

Lecture 2

LDAP

- LDAP (Lightweight Directory Access Protocol) is a standard, open protocol used to access and manage directory services over a network.
  - Think of a directory service as a centralized database (like a phone book or address book) that stores structured information about network resources—such as users, groups, computers, printers, and organizational units.
- LDAP allows applications, systems, and admins to query, add, modify, or delete this data efficiently.
- Microsoft follows the LDAP naming scheme within Active Directory to make its database compatible with other LDAP-compliant applications.
  - LDAP is how tools like PowerShell (via Get-ADUser) interact under the hood.
  - Enables data exchanges between directory services (such as Active Directory) and applications (Microsoft Exchange Server)

LDAP Distinguished Names

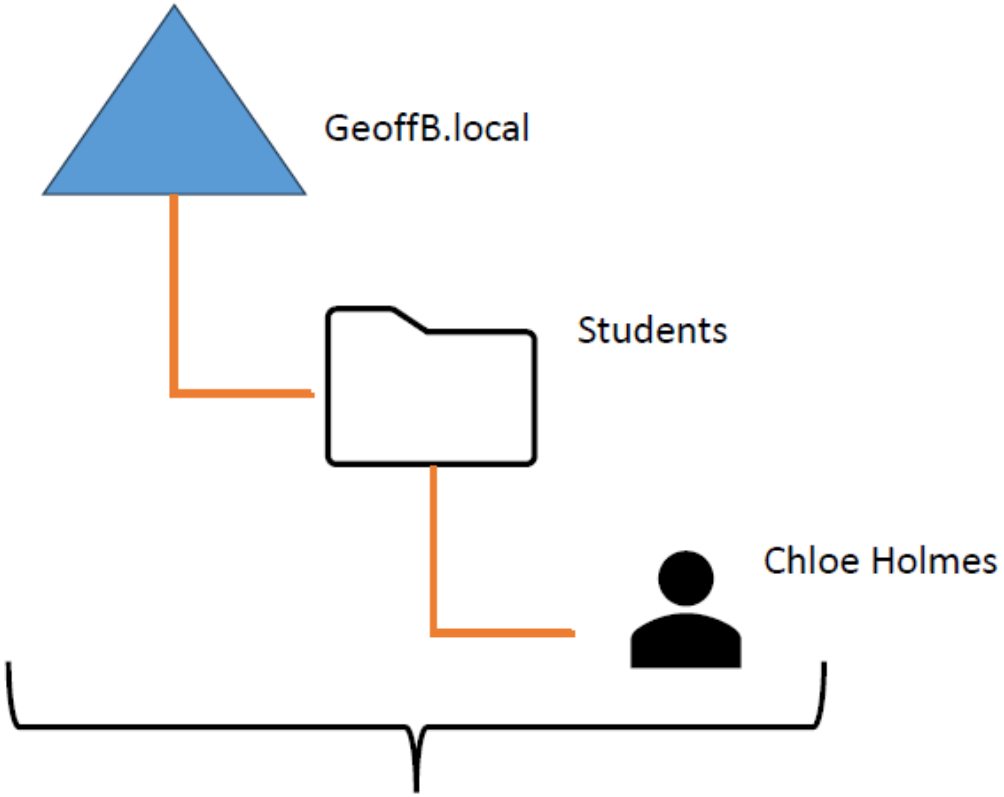
- The Distinguished Name (DN) is LDAP's naming convention for uniquely identifying an object in this directory tree—similar to a full file path on your computer (`e.g., C:\Users\John\Documents\file.txt`).
  - Structure: A DN is a comma-separated string of Relative Distinguished Names (RDNs), each representing a level in the hierarchy. It starts from the specific object and works up to the root.
    - Common RDN components:
      - CN (Common Name, e.g., any 'leaf' object: user, computer, group)
      - OU (Organizational Unit)
      - DC (Domain Component for DNS-like domains)
    - Example: CN=John Doe,OU=Sales,DC=contoso,DC=com
    - This means: User "John Doe" in the "Sales" OU, within the "contoso.com" domain.

LDAP Distinguished Names

- LDAP names are broken into sections which identify the path within the database to the object - LDAP names contain several keywords which help identify parts of an objects name:

Common RDNs	Description
CN	commonName
L	localityName
ST	stateOrProvinceName
O	organizationName
OU	organizationalUnitName
C	countryName
STREET	streetAddress
DC	domainComponent
UID	userid

Distinguished Name



CN=“Chloe Holmes”,OU=Students,DC=GeoffB,DC=Local

User Principal Name (UPN)

- Microsoft's own format for expressing usernames in a multi-domain environment
- Appears similar to an email address with the username@domainname
  - (jcadmin@clowes.local)
- Must be unique to each user account within the forest
- The UPN Suffix by default is the domain name, but can be configured to be other values as well

Pre-Windows 2000 (legacy) name

- Commonly referred to as the NetBIOS Domain Name
  - Used for backwards compatibility with older domain names
- Domain\samAccountName format
- Example:

GEOFFB\Administrator

Domain Name

Username Name
(samAccountName, or Logon Name)

Creating a Domain User Account

- To create domain user objects, you must be a member of the Enterprise Admins, Domain Admins, or Account Operators group, or you must have been delegated the administrative permissions.
- When creating a user, the Fullname and logon name are required parameters

User Account Management

- Default accounts:
  - Administrator, Default Account and Guest
- Accounts can be set up in two general environments:
  - Accounts that are set up through a stand-alone server that does not have Active Directory installed (local accounts; SAM database).
  - Accounts that are set up in a domain when Active Directory is installed (Domain accounts; Active Directory)
- When accounts are created in the domain through Active Directory, those accounts can be used to access any server or resource in the domain (they are permitted to).

User Account Naming Conventions

- A guideline for how network names will be generated
- Active Directory user logon names must be unique within enterprise
  - 20-character username maximum
  - Invalid characters: “ / \ [ ] : ; | = , + * ? < > @
- User logon names are **NOT** case sensitive
- Develop a strategy for employees with the same name

Creating Accounts When Active Directory Is Installed

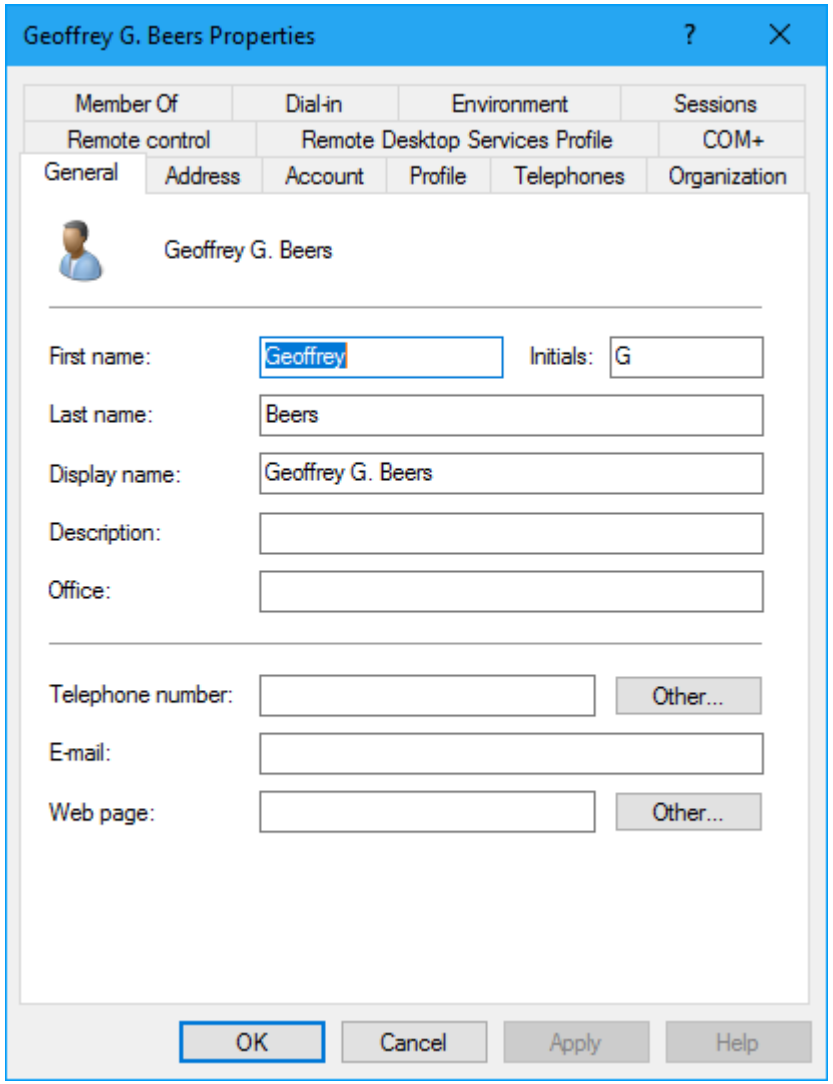
- When Active Directory *is* installed, and the server is a DC, use the Active Directory Users and Computers (ADUC) tool either from:
  - The Windows Administrative Tools folder (Start Menu)
  - Tools Drop down in Server Manager
  - MMC snap-in
  - Run → DSA.msc

Do this...

1. Create a user account with your name (first and last) and the login name of dadmin_<initals> (example: dadmin_ggb).
2. In the Address tab configured with the following:
    1. Street: 1234 Main Street
    2. City: Moncton
    3. State/Province: NB
    4. Postal Code: B4N-4N4
    5. Country: Canada
3. In the Account Tab set the following properties:
    1. Logon hours: 08:00 – 17:00, Monday to Friday.
    2. Account Expires: July 14th, 2025
    3. User password must be change when the user first logs on.

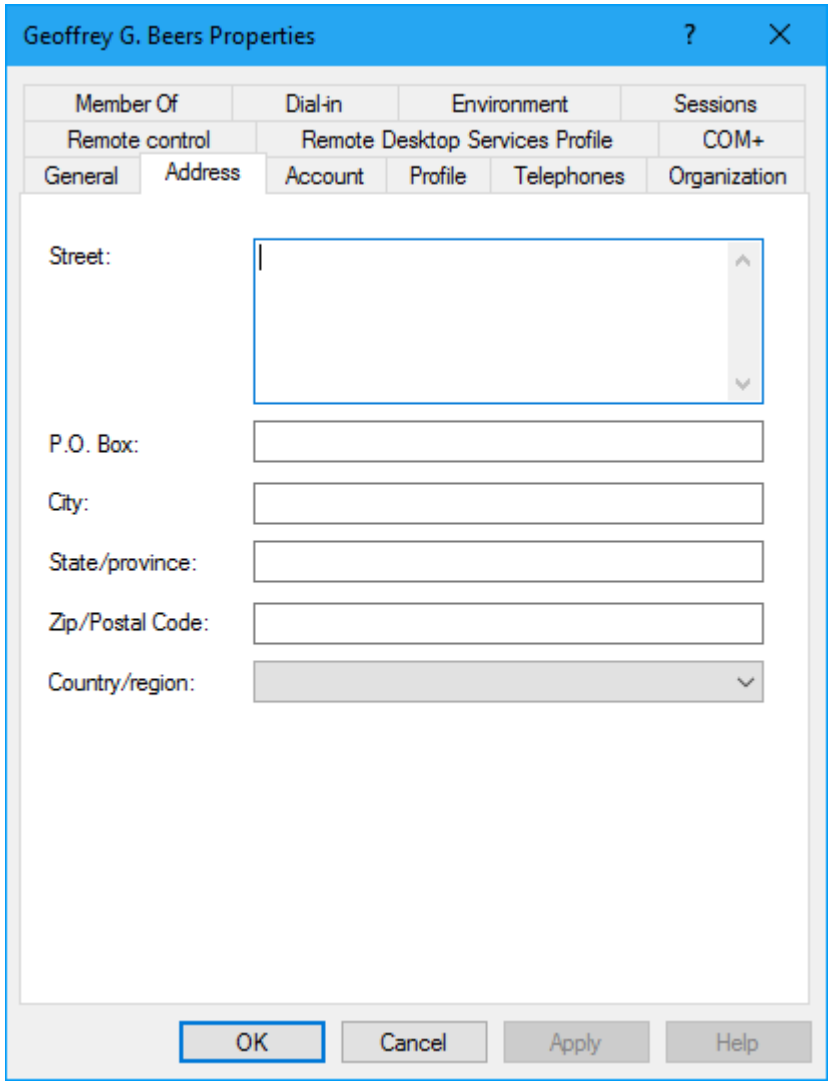
Examining User Accounts

User Account Properties



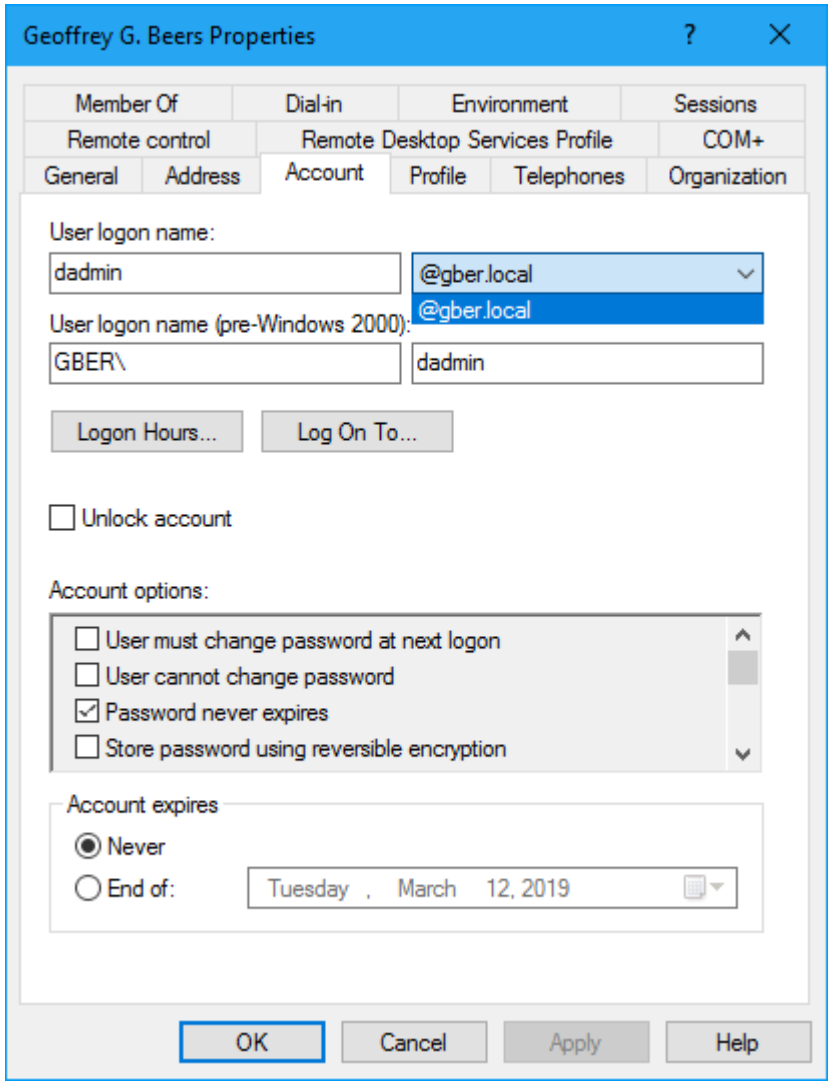
- **General tab**
  - Modify personal information about the user
  - Providing values on this tab enables administrators to locate domain user accounts by searching the directory for demographic information.

User Account Properties



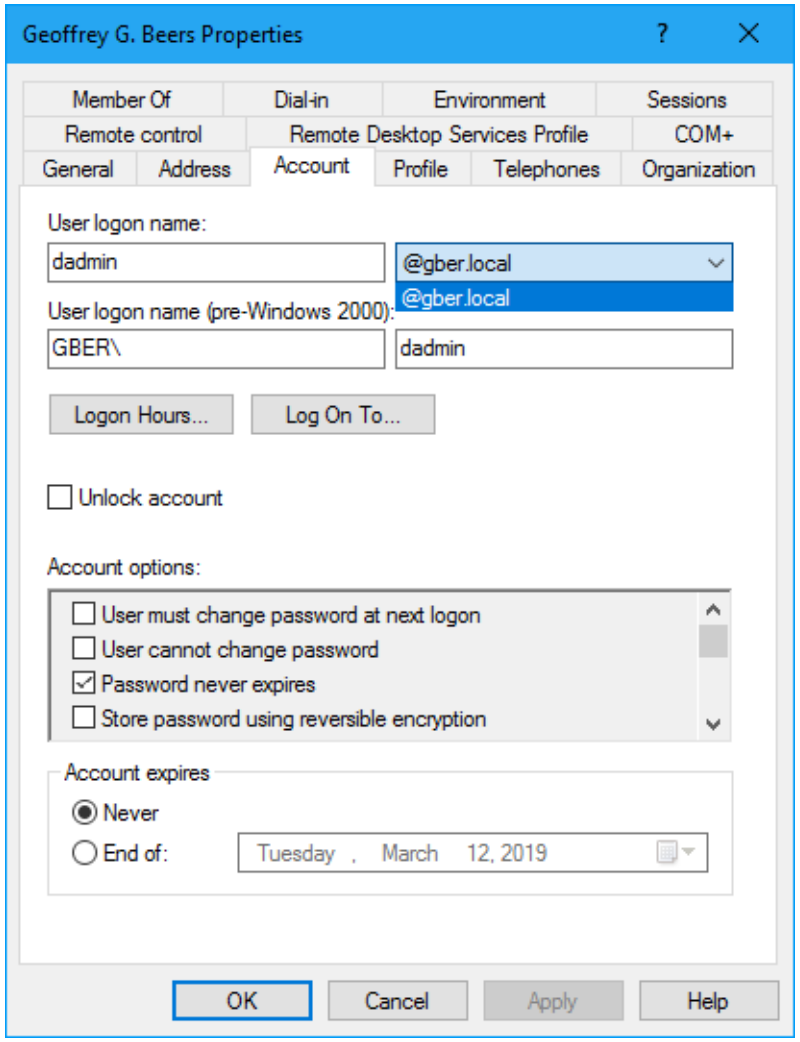
- **Address tab**
  - Provide more demographical information (the users physical location).

User Account Properties



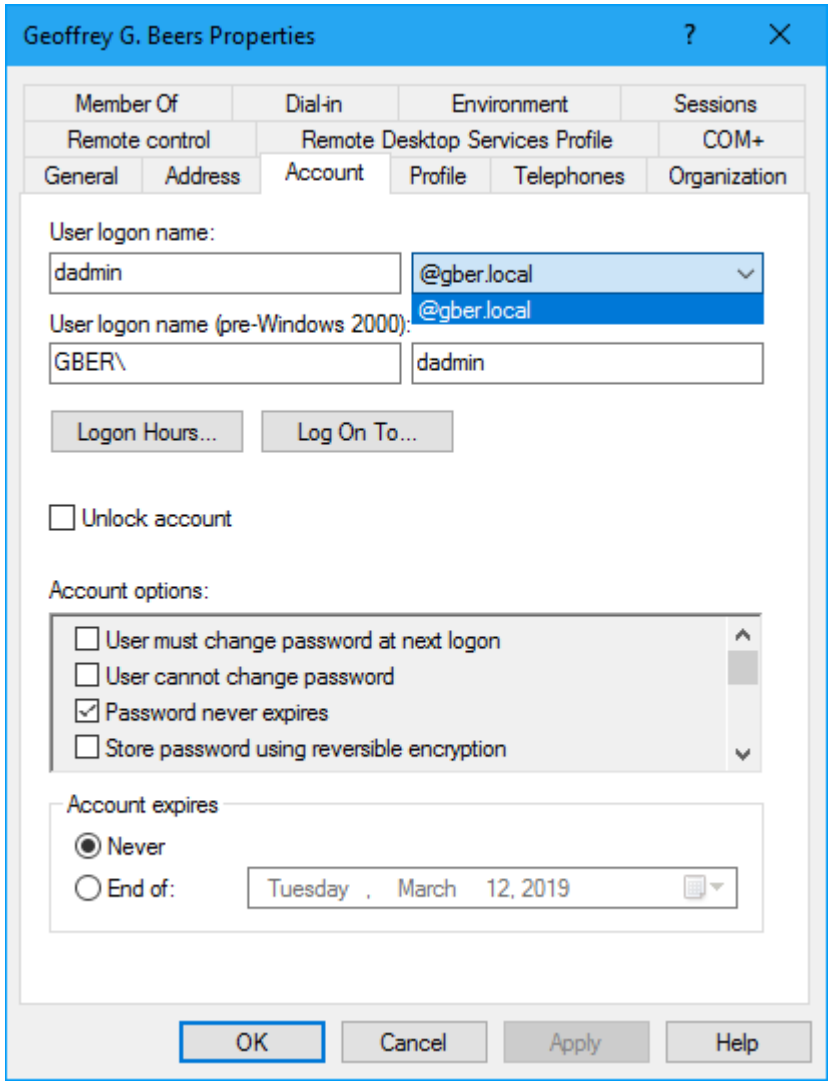
- **Account tab**
  - User logon name
    - Username in UPN format.
  - Pre-Windows 2000 logon name
    - Username in Legacy format.

User Account Properties



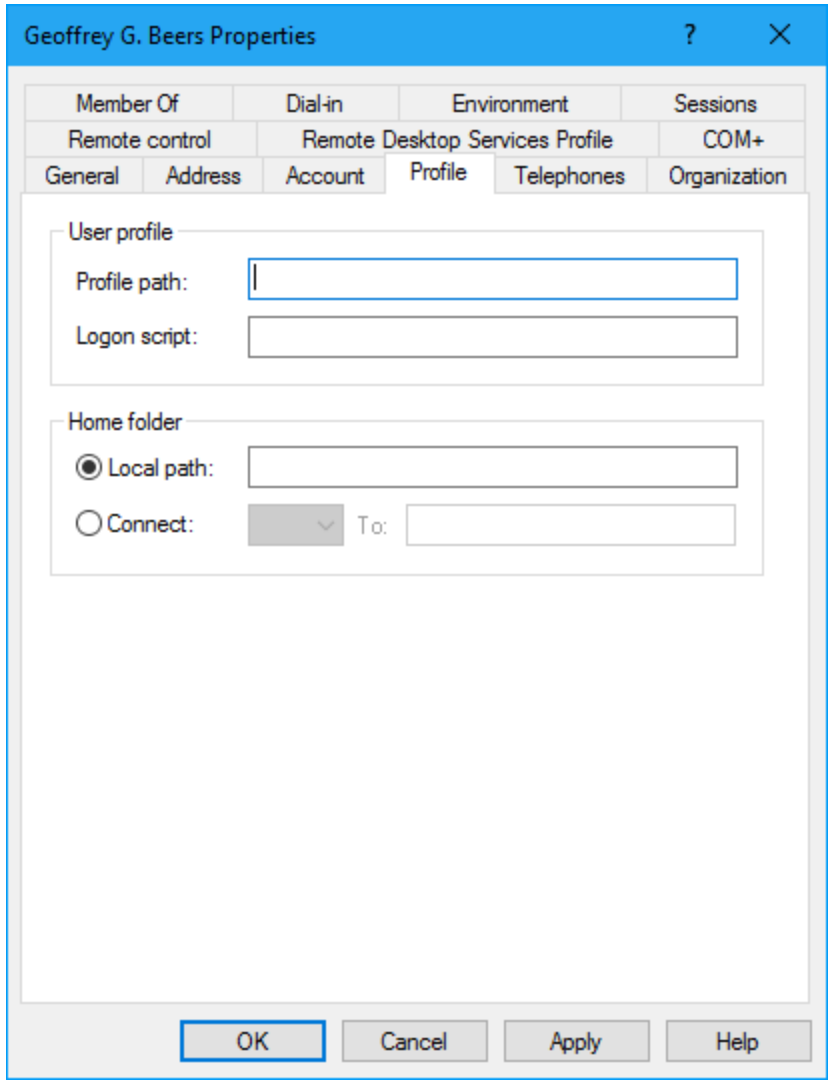
- **Account tab**
  - Logon Hours
    - which times of day and which days of the week the user is permitted to logon to the domain
    - only prevents a user from logging on, by default, doesn't interrupt an existing session
    - only applies to domain logons, not local logons

User Account Properties



- **Account tab**
  - Logon To
    - specifies the name of computers that this user can use to logon (restriction)
  - Unlock Account
    - disabled by default
    - activated when the user account is locked by too many failed logon attempts
  - Account Is Disabled
  - Account Expires
    - date on which the user account will automatically be disabled
  - User must change password at next logon.
    - Flags the users account to prompt the user for a password change at first logon

User Account Properties



- **Profile tab**
  - User profile is a collection of folders and data that stores your current desktop environment and application settings as well as personal data
    - Can be stored on a server so that it can follow the user from computer to computer. This is called a roaming user profile.
  - **Logon script**
    - A file that contains a series of commands to run each time a user logs onto his or her account, such as a command to map a home drive
    - Logon script has the extension .bat, .cmd, .vbs (Visual Basic script), or .js (Java script), .ps/ps1
  - **Home Folder**
    - A place to put your things
    - **Not** your profile path (i,e user folder).
    - Local path: a folder on the *local* computer
    - Connect:a mapped shared folder.

Assigning a Home Folder

- A home folder is an **additional folder** that is used by users to store personal documents
- **Steps to create a home folder:**
  - On a server, create and share a folder where you want the home folders to go
  - Give the User's group full control to the shared folder
  - Specify the path to the home folder in the user account's properties page, in the form \\<server>\<share>\<username> and choose a drive letter to map it to.
  - you can use the variable ***%username%*** in place of the username and Windows Server will automatically substitute the correct username
    - **DO NOT** use the %username% variable if this is being done in a script or CLI command.

Do this...

- Create a folder in the root of C: called UserShares
- Share UserShares (using advanced sharing) and give Everyone → Full Control share permission.
- Set the Home Folder → Connect to:
  - H:
  - \\Server1-XXX\UserShares\%username%

This is a variable for "whatever the current user's username is"

User Templates

- An object that is pre-populated with common properties.
- Accomplished by creating a generic user object and then copying that object to create new users.
- Configure the properties of the template that are to be common to all of the new users you create, including group membership.
- The template account should be disabled so no one can use it to logon with
- Permissions assigned to template objects are not copied to the object generated from the template

Properties Copied when Using a User Template

Tab	Properties Copied
General	None
Address	All, **except Street Address**
Telephones	None
Organization	All, except Title
Account	All, except User Logon Name and User Logon Name (pre-Windows 2000), which are specified during the copy process
Profile	All, including Profile Path and Local Path, which are modified to reflect the new user's logon name
Member Of	All
Terminal Services Profile, Environment, Remote Control, Sessions, Dial-In, COM+	None

Disabling, Enabling, and Renaming Accounts

- Why disable?
  - When a user takes a leave of absence
  - Security breach
- Organizations might also have the practice of disabling accounts when someone leaves and enabling the account for that person's replacement (for a transition period).
- Why Rename
  - Name Change
  - Configuration Error

Do this...

- Create a template from this account by copying it.
  - Create an OU for template accounts called: UserTemplates
  - Name the template account *regUsrTemp.
    - Place the account in the UserTemplates OU

Moving an Account

- When an employee moves from one department to another you might need to move that person’s account from one container to another.
- Remember:
  - Container = OU
  - Policies are applied to OUs
  - Therefore: moving to a new OU could/will change a person's **level of access**

Resetting a Password

- You do not have the option to look up a password, but you can reset it for users
- For organizations that have accounts that manage sensitive information
  - It is advisable to have specific guidelines that govern the circumstances under which an account password is reset

Do this…

- Using your DADMIN account:
  - Disable and Rename DADMIN account
  - Reset the password for the account
- Use Powershell
  - `Get-ADUser –Identity <samAccountName> -Properties ObjectGUID`

  …View the GUID/SID for the account.

Deleting an Account

- It is a good practice to delete accounts that are no longer in use (at the very least, disable them)
  - Most organizations will have a retention policy for the length of time accounts should be kept.
- Failure to do so could expose your company to security risks
- When you delete an account its globally unique identifier (GUID) is also deleted and will not be reused even if you create another account using the same name.
  - User SIDs are unique to the Domain
  - GUIDs are unique…*full stop*